Journal of Thermal Analysis and Calorimetry (2021) 144:1343–1349
https://doi.org/10.1007/s10973-020-09558-2

Thermal analysis of the oxide–chloride systems $GdCl_3$–Gd_2O_3 and $GdCl_3$–KCl–Gd_2O_3

Iraida V. Korzun[1] · Elena V. Nikolaeva[1,2] · Irina D. Zakir'yanova[1,2]

Received: 9 August 2019 / Accepted: 13 March 2020 / Published online: 13 April 2020
© Akadémiai Kiadó, Budapest, Hungary 2020

Abstract
The thermal analysis of the oxide–chloride systems $GdCl_3$–Gd_2O_3 and $GdCl_3$–KCl–Gd_2O_3 with the Gd_2O_3 content up to 0.09 mol fractions was studied in the temperature range 298–1173 K using two methods: differential scanning calorimetry and the method of cooling curves. The liquidus temperatures in these systems have been determined. The solubility of the oxide and the thermodynamic characteristics of the dissolution process were calculated. It was established experimentally that the dissolution of Gd_2O_3 proceeds by a chemical mechanism, with the formation of GdOCl oxychloride. The received data revealed the main regularities of Gd_2O_3 solubility in molten $GdCl_3$–KCl–Gd_2O_3 systems.

Keywords Gadolinium chloride · Gadolinium oxide · DSC · Method of cooling curves · Liquidus temperatures · Solubility

Introduction

The application of rare earth metals (REM) in technology and science is expanding every year. In particular, gadolinium and its compounds are actively used in nuclear energy, electronics and medicine.

The most important methods of the rare earth metal production are the electrolysis of their anhydrous chlorides in molten alkali metal chloride electrolytes and the metallothermy in molten chloride and fluoride systems. [1]. In the process of REM chlorides production, the direct conversion of oxide natural raw materials with chlorinating agents is used [2].

Now approaches are developed for the immediate electrochemical reduction of solid metal oxides into their metals in molten salts. This method has been extensively adopted in metal extraction for a number of elements including REM (Gd, Tb, Dy, Er, and Ce) [3–5].

In fact, the physicochemical properties of the oxide–chloride systems $GdCl_3$–Gd_2O_3 and their mixtures with alkali metal chlorides have not been studied. Data for liquidus temperatures are not available. The solubility of Gd_2O_3 in these melts is unknown.

The phase diagrams of similar binary systems Re_2O_3–$ReCl_3$ (Re = La, Nd, Y), published in [6, 7], are doubtful, since investigations of these chemically aggressive oxide–chloride melts were carried out in quartz vessels. Because of possible impurities, the melting points of rare-earth chlorides are lower than in the later literature.

This paper presents the results of thermal analysis of oxide–chloride systems $GdCl_3$–Gd_2O_3 and $GdCl_3$–KCl–Gd_2O_3, including the determination of liquidus temperatures, solubility of gadolinium oxide in chloride melts and thermodynamic characteristics of dissolution. The oxide–chloride systems $GdCl_3$–Gd_2O_3 and $GdCl_3$–KCl–Gd_2O_3 with the Gd_2O_3 content up to 0.09 mol fractions were investigated.

Experimental

Preparation of reagents and their certification

The Gd_2O_3 (extra pure grade; "Vekton," St. Petersburg, Russia) powder was dried at 973 K for 4 h. X-ray diffraction and Raman spectroscopy (Fig. 1) confirmed the monophasic nature of the obtained product. In the registered Raman spectrum of Gd_2O_3, a set of vibration frequencies

✉ Iraida V. Korzun
korzun@ihte.uran.ru

[1] Institute of High Temperature Electrochemistry, Ural Branch of the Russian Academy of Sciences, Akademicheskaya St., 20, Yekaterinburg, Russian Federation 620137

[2] Ural Federal University, Mira St., 19, Yekaterinburg, Russian Federation 620002

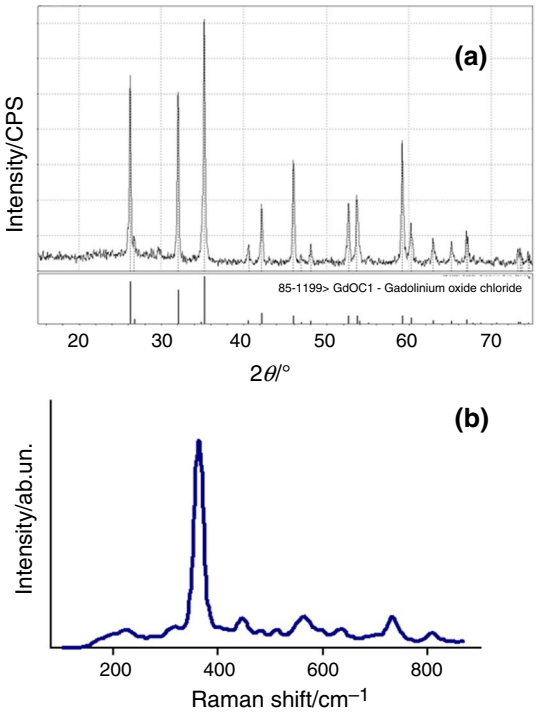


Fig. 1 X-ray diffraction pattern of Gd_2O_3 (**a**); Raman spectra of Gd_2O_3 (**b**)

is observed (226, 315, 363, 445, 563 cm^{-1}), which, in accordance with [8], corresponds to the cubic modification of gadolinium oxide (C-modification).

Vibration bands corresponding to any impurities were not detected.

Gadolinium trichloride $GdCl_3$ was obtained in two stages.

1. Gd_2O_3 powder was placed in a glass-carbon cup and dissolved in an excess of HCl:

$$Gd_2O_3 + HCl \rightarrow GdCl_3 \cdot 6H_2O + H_2O \uparrow \quad (1)$$

$GdCl_3{\cdot}6H_2O$ was obtained by evaporation of the solution, while it was slowly heated to temperatures not exceeding 100° C.

2. Anhydrous $GdCl_3$ was obtained from its hydrate, $GdCl_3{\cdot}6H_2O$. In our previous work [9], it was convincingly proved that the dehydration of the gadolinium chloride hydrate when heated in an inert atmosphere is accompanied by hydrolysis processes. To prevent this, anhydrous gadolinium trichloride was obtained by chlorinating crystalline hydrate $GdCl_3{\cdot}6H_2O$ with heating in a stream of high-purity argon saturated with carbon tetrachloride (CCl_4) vapor at 823 K for 24 h, as described in [10].

Potassium chloride (chemically pure grade; "Vekton," St. Petersburg, Russia) was melted in an argon atmosphere and then purified by zone melting method.

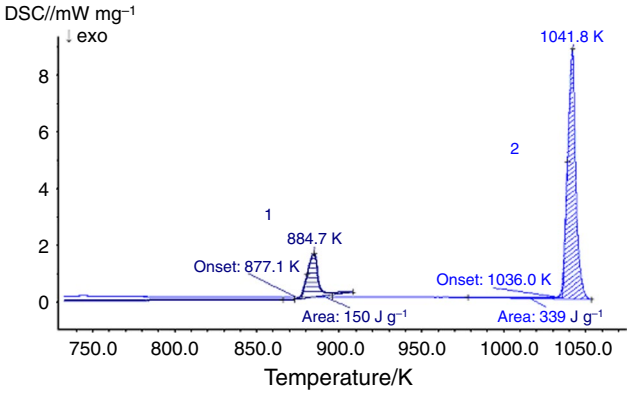


Fig. 2 DSC curves for $GdCl_3$ (1) and KCl (2)

The synthesized anhydrous $GdCl_3$ and KCl are certified by the method of simultaneous thermal analysis (Fig. 2). The melting point of $GdCl_3$, measured by various authors [11–16], varies over a rather wide range, from 873 to 884 K. The melting enthalpy, accepted in the literature [14], is about 40 kJ mol^{-1}. For potassium chloride, the melting point and enthalpy are 1044 K and 26.3 kJ mol^{-1} [17]. Our results are consistent with published data.

To prepare the mixtures, the KCl and $GdCl_3$ samples weighted in appropriate proportions were melted in a glassy carbon crucible and kept at 1093 K for 5 h in an argon atmosphere. All operations with chemical reagents and mixtures were carried out in a glove box with a nitrogen atmosphere. Chloride systems ($0.69GdCl_3$–0.31KCl and $0.543GdCl_3$–0.457KCl) close to the eutectic compositions [18] were chosen for the investigation, allowing the experiments to be carried out in a relatively wide temperature range.

Experimental

The liquidus temperatures were obtained by two methods: differential scanning calorimetry (DSC) and the method of cooling curves. DSC studies were performed using a thermal analyzer STA 449C Jupiter® (NETZSCH) with a heating rate of 10 K min^{-1} in a high-purity argon atmosphere in Pt–Rh crucibles with punches in the covers. According to the technical parameters of the experimental set, the temperature detection error did not exceed 1 K. Additionally, thermogravimetric measurements showed that the reduction in sample mass due to evaporation did not exceed 1.5 mass%.

According to Ref. [19], two or three measurements were performed under the same conditions for each composition. The samples were not removed from the device. During a primary heating, the Gd_2O_3 interacted with the chloride melt. The secondary experiments were begun after cooling

the device to the room temperature. The results of the secondary observations were used to determine the liquidus temperature. The third measurements were carried out to ensure the reproducibility of the results.

For KCl and $GdCl_3$, the maximum temperatures were not more than 50 degrees higher their melting points. For the KCl-GdCl systems, the maximum temperatures were selected in line with their liquidus temperatures according to phase diagram [18]. For the oxide–chloride systems, the maximum temperatures were not more than 1000 K. At higher temperatures, the decomposition of GdOCl should begin according to our previous research [20].

The method of cooling curves was described in detail in our previous works [21, 22]. The measurements were performed in inert argon atmosphere. The composition under study was placed into the glassy carbon crucible. The hermetical quartz container with the crucible inside was inserted into a massive metallic block of the electric furnace to provide the required cooling rate (2–3° min^{-1}). The temperature in the furnace was controlled by a microprocessing thermoregulator VARTA TP-403 and recorded by a Pt/PtRh thermocouple using a digital multimeter APPA 109 N with a frequency of 2 measurement per second. Temperature was measured by means of Pt-PtRh thermocouple. Each thermocouple was checked periodically at the freezing point of aluminum or zinc. The melt was reheated 50–70 °C higher than the temperature of the supposed primary melt crystallization and kept under those conditions for a few hours to reach the equilibrium state. The measurements were carried out during the period of the melt cooling down. Cooling curves in temperature–time coordinates were automatically recorded. Several determinations were usually made for each composition. The total error of the liquidus temperature measurement didn't exceed 5°.

With a significant release of crystallization energy, a bend (decrease in the cooling rate) which corresponds to the beginning of crystallization of the more refractory component is observed on the cooling curve. The bend on the cooling curve can be unobtrusive if the crystallization energy release is negligible. In such a case, the data were received by mathematical treatment of the cooling curve using the Table Curve 2D v.5.01 software. The results agree well with the DSC measurement data.

As an example, the DSC curve and the cooling curve obtained for the system (0.543GdCl3–0.457KCl) are shown in Fig. 3. It is seen that the solidus (768.7 K) and liquidus (823.2 K) temperatures, measured by the DSC method, are in good agreement with the results obtained by the method of cooling curves (772 K for solidus and 826 K for liquidus temperature).

The Gd_2O_3 additives to the melt cause changes in the experimental curves (Fig. 4).

The Raman spectra were recorded using the optic fiber spectrometric Ava-Raman complex (Avantes, The Netherlands), which includes the monochromatic laser (λ = 532 nm, radiation power of 50 mW). Spectra were recorded using a 180° optical scattering arrangement.

The phase compositions of the reagents used were determined using a Rigaku D/MAX-2200VL/PC X-ray diffractometer (Japan).

Results and discussion

Liquidus temperatures of the oxide–chloride systems $GdCl_3$–Gd_2O_3, with the content of gadolinium oxide up to 0.09 mol fractions

The experimental data for binary oxide–chloride system $GdCl_3$–Gd_2O_3 with additions up to 0.09 mol fractions of Gd_2O_3 obtained by the DSC method are shown in Fig. 5. It can be seen that in the investigated range of Gd_2O_3

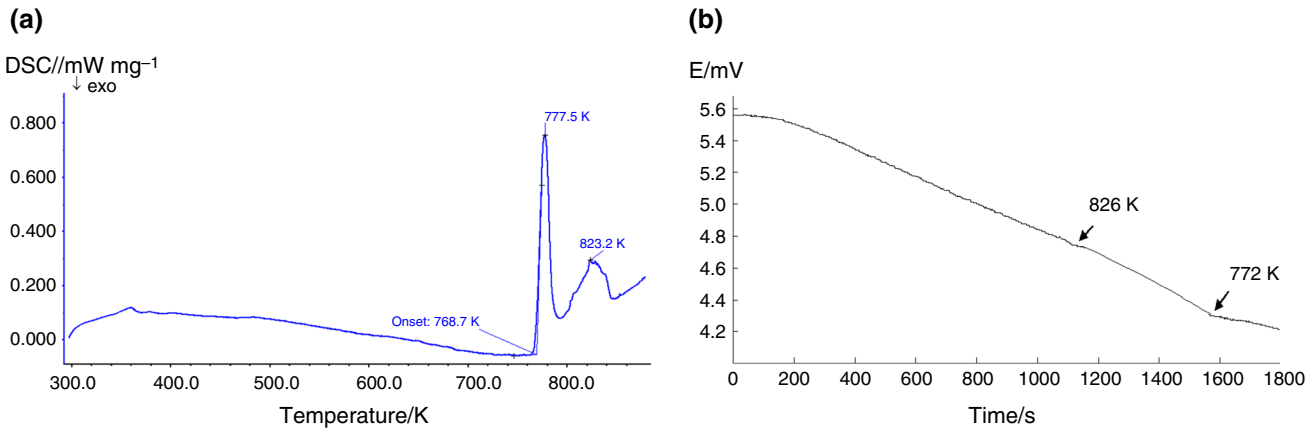


Fig. 3 The system 0.543GdCl3–0.457KCl: **a** DSC curve; cooling curve (**b**)

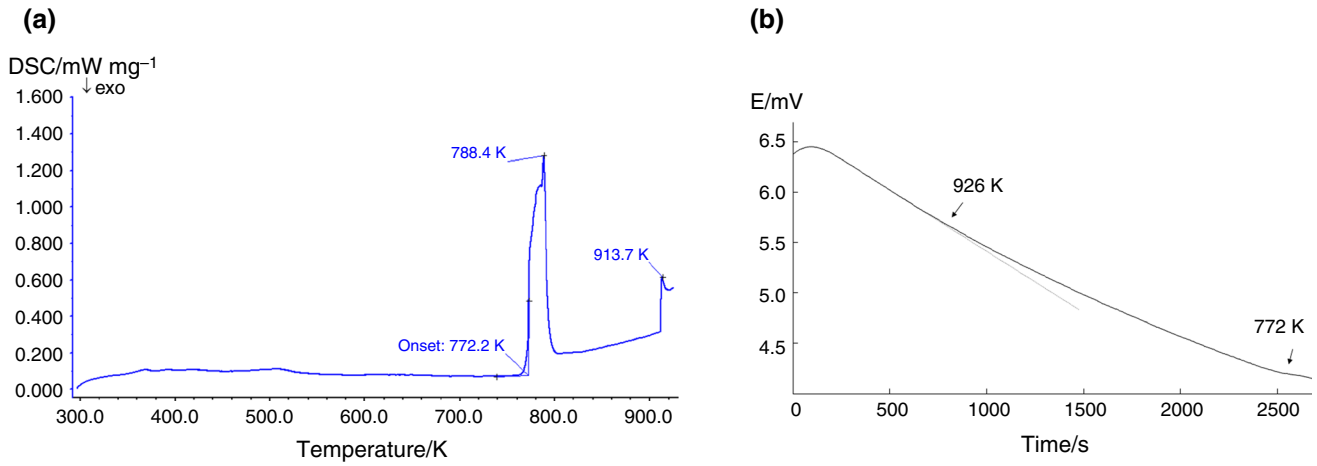


Fig. 4 The system 0.543GdCl3–0.457KCl–Gd_2O_3: **a** DSC curve for (0.543$GdCl_3$–0.457KCl)–0.0125Gd_2O_3 (the secondary measurement); **b** cooling curve for (0.543$GdCl_3$–0.457KCl)–0.0153 Gd_2O_3

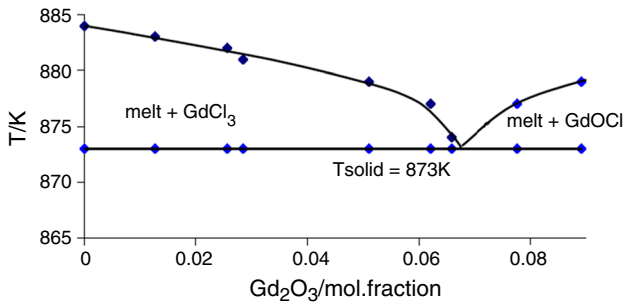


Fig. 5 Fragment of the phase diagram of the binary oxide–chloride system $GdCl_3$–Gd_2O_3

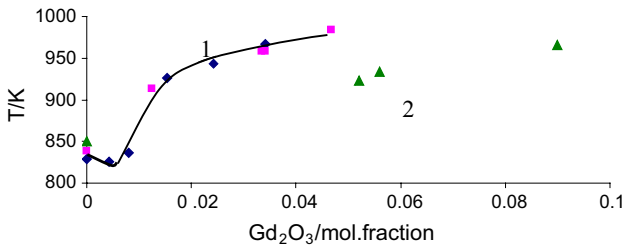


Fig. 6 Liquidus points for (1) (0.543$GdCl_3$–0.457KCl)–Gd_2O_3: pink filled box DSC, blue filled diamond method of cooling curves; (2) (0.69$GdCl_3$–0.31KC)–Gd_2O_3: (green filled triangle DSC). (Color figure online)

concentrations experimental points fit with the eutectic type diagram. The eutectic point was established as an intersection of the liquidus and the solidus curves. The eutectic point coordinates are N (Gd_2O3) = 0.068 and T = 873 K.

Liquidus temperatures of chloride–oxide systems Gd_2O_3–$GdCl_3$–KCl, with the content of gadolinium oxide up to 0.09 mol fractions

Figure 6 demonstrates liquidus temperatures for 0.543$GdCl_3$–0.457KCl and 0.69$GdCl_3$–0.31KCl chloride systems with Gd_2O_3 additives.

The addition of gadolinium oxide to the chloride system at first slightly decreases and then increases the temperature of the primary crystallization. The minimum on the liquidus curve for the (0.543$GdCl_3$–0.457KCl)–Gd_2O_3 system is observed at N(Gd_2O_3) = 0.005 and T = 826 K.

The composition of the chloride mixture affects the liquidus temperature of the oxide–salt mixture. In particular, an increase in the content of gadolinium chloride leads to a decrease in the liquidus temperature (Fig. 6): the liquidus temperatures of (0.543$GdCl_3$–0.457KCl)–Gd_2O_3 system are higher than those of (0.69$GdCl_3$–0.31KCl)–Gd_2O_3 system at the same Gd_2O_3 concentrations.

The liquidus temperatures for all the systems studied are summarized in Table 1.

Thermodynamic parameters of dissolution of gadolinium oxide in melts $GdCl_3$ and $GdCl_3$–KCl

The dependences of liquidus temperatures on the added gadolinium oxide concentration are shown in Figs. 5 and 6. At first, the Gd_2O_3 additions decrease the liquidus temperature and then increase it. The change of the dependence flow may be due to the new solid phase which was the first to crystallize during the melt cooling. For the present instance, it should be the GdOCl. The coordinates of the points on the ascending branches of the liquidus curves correspond to the solubility of Gd_2O_3 (S) in the melt. The solubility

Table 1 Liquidus temperatures (T_{liq}) of Gd_2O_3–$GdCl_3$ and Gd_2O_3–$GdCl_3$–KCl oxide–chloride systems

System	N (Gd_2O_3)/m.f.	T_{liq}/K
$GdCl_3$–Gd_2O_3	0	884
	0.0128	883
	0.0257	882
	0.0285	881
	0.051	879
	0.062	877
	0.0657	874
	0.0775	877
	0.089	880
(0.543$GdCl_3$–0.457KCl)–Gd_2O_3	0	829
	0.0042	826
	0.0125	914
	0.0153	926
	0.0243	943
	0.0335	959
	0.0341	967
	0.0467	984
(0.69$GdCl_3$–0.31KCl)–Gd_2O_3	0	850
	0.052	923
	0.056	934
	0.09	966
	0.0257	882
	0.0285	881

values thus obtained may be approximated by Eq. (2) in the temperature range under study:

$$\ln S = A + B/T, \tag{2}$$

where A and B are the empirical coefficients.

Table 2 shows the values of A and B for the systems under study and the values of Gd_2O_3 solubility of at certain temperatures.

Figure 7 shows the temperature dependence of the solubility of Gd_2O_3 in the studied solvents.

(ln$S$ vs. $1/T$). It is obvious that solubility of gadolinium oxide increases with increasing content of $GdCl_3$ in the system. This fact agrees well with the literature data [6, 23].

The Gibbs energy changes during the Gd_2O_3 dissolution can be expressed by (3).

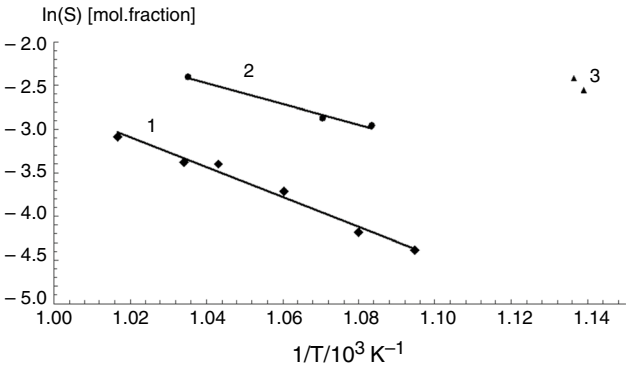


Fig. 7 Temperature dependence of Gd_2O_3 solubility in molten chloride systems: (1) 0.543$GdCl_3$–0.457KCl; (2) 0.69$GdCl_3$–0.31KCl и (3) $GdCl_3$

$$\Delta\bar{G}^* = -RT\ln k^* = -RT\ln S, \tag{3}$$

where $k^* = k/[\gamma(Gd_2O_3)]$ is the conditional equilibrium constant of the reaction: Gd_2O_3 (solid) ↔ Gd_2O_3 (solution), with the assumption that the gadolinium oxide activity coefficients γ (Gd_2O_3) in chloride are unchangeable in the concentration range under study.

Substitution of (2) to (3) gives the following equation for temperature dependence of the Gibbs energy changes:

$$\Delta\bar{G}^* = \Delta\bar{H}^* - T\Delta\bar{S}^* = -RT\left(A + \frac{B}{T}\right) \tag{4}$$

The conditional enthalpy ($\Delta H^* = -RB$) and entropy ($\Delta S^* = RA$) changes during the Gd_2O_3 dissolution are presented in Table 3.

Positive enthalpy values of oxide dissolution indicate that this process is accompanied by heat absorption. One can see that at a constant temperature ΔG^* decreases with increasing concentration of $GdCl_3$ in the melt-solvent, and the endothermic effect of dissolution also decreases. This fact indicates a reduction in the energy consumption of the dissolution process, which may be due to the exothermic interaction of gadolinium oxide with $GdCl_3$ during their mixing. The concentration of $GdCl_3$ is higher in 0.69$GdCl_3$–0.31KCl than in 0.543$GdCl_3$–0.457KCl system, and properly the conditional enthalpy of Gd_2O_3 dissolution in the first system is lower.

Table 2 Coefficients of (2) for the molten $GdCl_3$ and $GdCl_3$–KCl systems and the values of Gd_2O_3 solubility (S)

Solvent	Temperature interval/K	A	$-B\times10^4$	$\pm\Delta(\ln S)$	S/mole fraction	
					950 K	880 K
0.543$GdCl_3$–0.457KCl	836–984	14.37	1.71	0.08	0.026	0.006
0.69$GdCl_3$–0.31KCl	923–966	9.80	1.18	0.10	0.072	0.027
$GdCl_3$	880	–	–	–	–	0.089

Table 3 Thermodynamic characteristics of the Gd_2O_3 solubility in molten chlorides

Solvent	ΔH*/kJ mol^{-1}	ΔS*/J mol^{-1} K^{-1}	ΔG*/kJ mol^{-1}		
			880K	930 K	950 K
0.543$GdCl_3$–0.457KCl	142.36	119.6	37.21	31.24	28.85
0.69$GdCl_3$–0.31KCl	98.15	81.5	26.45	22.36	20.73
$GdCl_3$	–	–	17.70	–	–

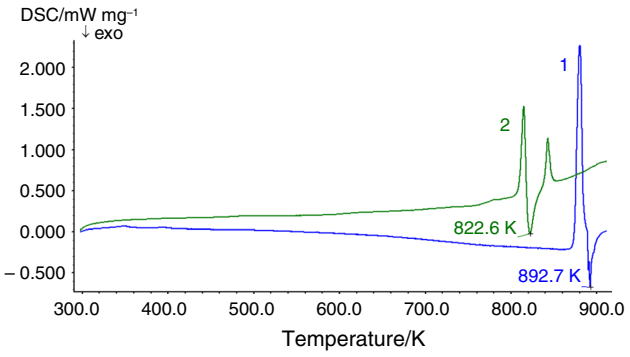


Fig. 8 Primary DSC measurement results: (1) $GdCl_3$ with an addition 0.05 mol fr. Gd_2O_3; (2) 0.69$GdCl_3$–0.31KCl with 0.09 mol fr. Gd_2O_3

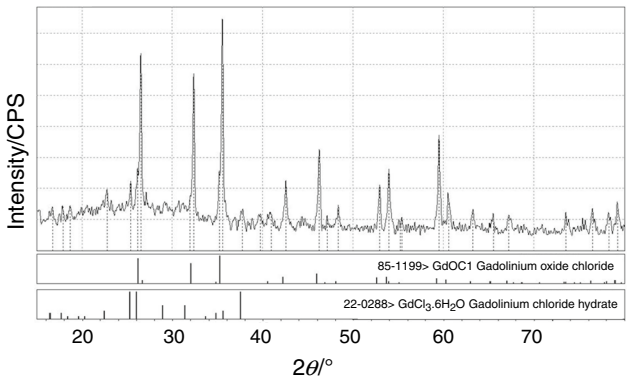


Fig. 9 X-ray diffraction pattern of frozen melt $GdCl_3$–Gd_2O_3 after experiment

When Gd_2O_3 was added to gadolinium chloride or a $GdCl_3$–KCl mixture, the DSC curves (Fig. 8) demonstrated the endothermic peaks corresponding to the $GdCl_3$ (curve 1) or chloride mixture (curve 2) melting and then the abrupt peak of exothermic effect. The appearance of exothermic peak testified the chemical reaction between gadolinium oxide and chloride melt. This fact confirms the thermodynamic calculations and provides direct evidence of the chemical reaction of the interaction between Gd_2O_3 and $GdCl_3$: $GdCl_3 + Gd_2O_3 = 3GdOCl$. The second endothermic peak on curve 2 should be attributed to melting of the $GdCl_3$–KCl–GdOCl composition.

For all the solvents under study, the XRD data of frozen melts after the experiment show the presence of GdOCl (Fig. 9).

In addition, Raman in situ spectroscopic analysis provided direct evidence of the chemical interaction between Gd_2O_3 and chloride melts containing $GdCl_3$. Indeed, when cooling a homogeneous melt $GdCl_3$–KCl–Gd_2O_3 below the liquidus temperature, in the Raman spectrum intense bands appear at 370 and 499 cm^{-1} (Fig. 10 spectrum 1), the position and relative intensities of which are consistent with the corresponding data for solid GdOCl [24]. The spectrum of the frozen melt (Fig. 10 spectrum 2) contains bands 383 and 511 cm^{-1} associated with the GdOCl phase. The characteristic vibrational bands of gadolinium oxide were not found.

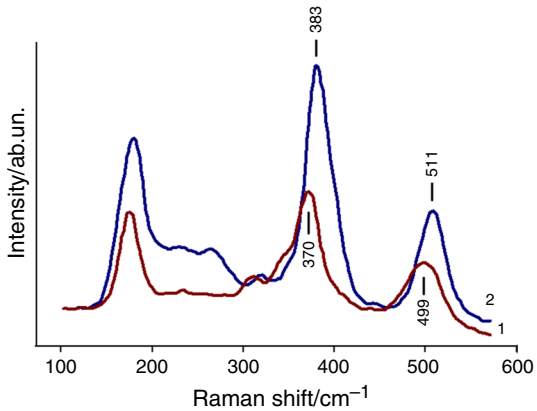


Fig. 10 Raman spectra of the 0.515$GdCl_3$–0.485KCl heterogeneous system with the addition of 0.07 mol fr. Gd_2O_3: 973 K (1); Raman spectrum of frozen melt (297 K) (2)

Conclusion

This paper provides new information on liquidus temperatures of $GdCl_3$–Gd_2O_3, (0.69$GdCl_3$–0.31KCl)–Gd_2O_3 and (0.543$GdCl_3$–0.457KCl)–Gd_2O_3 systems with Gd_2O_3 concentrations up to 0.09 mol fractions, determined by the thermal analysis methods. The addition of gadolinium oxide to the chloride system at first slightly decreases and then increases the temperature of the primary

crystallization. The increase in the gadolinium chloride content leads to the liquidus temperature reduction at the same Gd_2O_3 concentration.

Some regularities of Gd_2O_3 dissolution in molten $GdCl_3$ and $GdCl_3$–KCl systems were revealed. The gadolinium oxide solubility increases as temperature and $GdCl_3$ content increases. The thermodynamic characteristics of the gadolinium oxide dissolution in chloride melts were calculated.

The exothermic effects fixated at primary DSC measurement of oxide chloride mixtures in addition to the Raman spectroscopy results provided direct evidence of chemical interaction between Gd_2O_3 and chloride melts with the gadolinium oxychloride GdOCl formation.

Acknowledgements The partial financial support of the Russian Foundation for Basic Research, Project № 18-03-00561a.

References

1. McCarthy GJ, Rhyne JJ, Silber HB. The rare earth in modern science and technology, vol. 2. New York: Springer; 1980. https://doi.org/10.1007/978-1-4613-3054-7.
2. McCarthy GJ, Rhyne JJ, Silber HB, editors. The rare earth in modern science and technology, vol. 2. New York: Plenum Press; 1980.
3. Hirota K. Electrochemical deoxidation of RE–O (RE = Gd, Tb, Dy, Er) solid solutions. J Alloy Compd. 1999;282:101–8.
4. Claux B, Serp J, Fouletier J. Electrochemical reduction of cerium oxide into metal. Electrochim Acta. 2011;56:2771–80.
5. Wang D. Electrochemical metallization of solid terbium oxide. Angew Chem Int Ed. 2006;45:2384–8.
6. Baev AE, Novikov GI. Thermodynamic study of rare earth oxychlorides. Zh Neorg Khim. 1969;10(11):2457–64.
7. Tyen FI, Morozov IS. Interaction of rare earth oxychlorides with their chlorides in the melt. Zh Neorg Khim. 1969;14:2246–52.
8. Jinqiu YU, Lei CUI, Huaqiang HE, Shihong YAN, Yunsheng HU, Hao WU. Raman spectra of RE_2O_3 (RE = Eu, Gd, Dy, Ho, Er, Tm, Yb, Lu, Sc and Y): laser-excited luminescence and trace impurity analysis. J Rare Earths. 2014;32:1–4. https://doi.org/10.1016/S1002-0721(14)60025-9.
9. Korzun IV, Zakir'yanova ID, Nikolaeva EV. Mechanism and caloric effects of the thermal dehydration of $GdCl_3 \cdot 6H_2O$ crystalline hydrate. Russ Metall (Metall). 2018;8:722–7. https://doi.org/10.1134/s0036029518080104.
10. Revzin G.E. Anhydrous chlorides of rare earth elements and scandium. [Bezvodnye hloridy redkozemelnyh elementov i skandia]. In: Methods of obtaining of chemical reagents and preparations [Metody polucheniya himicheskih reaktivov i preparatov]. M.: IREA. 1967. 129 p. **[In Rus.]**.
11. Barin I, Kubaschewski O. Thermochemical properties of inorganic substances. Berlin: Springer; 1977. https://doi.org/10.1002//bbpc.197800023.
12. Barin I. Thermochemical data of pure substances. Weinheim: VCH Verlags Gesellschaf; 1989. https://doi.org/10.1002/ange.19901020738.
13. Seifert HJ. Melting points of lanthanide trichlorides: an unsolved problem. J Therm Anal Calorim. 2005;82:575–80. https://doi.org/10.1007/s10973-005-6946-7.
14. Rycerz L, Gaune-Escard M. Lanthanide (III) halides: thermodynamic properties and their correlation with crystal structure. J Alloy Compd. 2008;450:167–74. https://doi.org/10.1016/j.jallcom.2006.12.096.
15. Konings RJM, Kovács A. Thermodynamic properties of the lanthanide (III) halides. Chapter 213. In: Gschneidner KA, Bunzli J-C, Pecharsky V, editors. Handbook on physics and chemistry of rare earths, vol. 33. Amsterdam: Elsevier; 2003.
16. Brown D. Halides of the lanthanides and actinides. London: Wiley; 1968.
17. Glushko VP, editor. Thermodynamic properties of individual substances, vol. 4. Moscow: Nauka; 1982.
18. Ma Zhisen, Sun Yimin, Yan Ding Yu, Wang Zhiyu Qiao, Ye Xinyu. Thermodynamic calculation of the GdCl3–ACl (A = Na, K, Rb, Cs) phase diagrams based on experimental data. Comput Coupling Phase Diagr Thermochem. 2006;30:88–94.
19. Rycerz L. Practical remarks concerning phase diagrams determination on the basis of differential scanning calorimetry measurements. J Therm Anal Calorim. 2013;113:231–8. https://doi.org/10.1007/s10973-013-3097-0.
20. I.D. Zakir'yanova, I.V. Korzun. Thermal stability of oxichlorides LnOCl (Ln = Gd, Yb). In: Book of abstracts XXII International conference on chemical thermodynamics in Russia (RCCT 2019) Saint Petersburg, Russia, June 19–23, 2019, p. 287.
21. Nikolaeva EV, Zakirianova ID, Korzun IV, Bovet AL, Antonov BD. Interaction between barium oxide and barium containing chloride melt. Z Naturforsch. 2015;70(5):325–31. https://doi.org/10.1515/zna-2014-0370.
22. Nikolaeva EV, Zakiryanova ID, Bovet AL, Korzun IV. On barium oxide solubility in barium-containing chloride melts. Z Naturforsch. 2016;71(8):731–4. https://doi.org/10.1515/zna-2016-0163.
23. Castrillejo Y, Bermejo MR, Barrado E, Martinez AM, Diaz Arocas P. Solubilization of rare earth oxides in the eutectic LiCl–KCl mixture at 450 C and in the equimolar CaCl2–NaCl melt at 550 C. J Electroanal Chem. 2003;545:141–57. https://doi.org/10.1016/S0022-0728(03)00092-5.
24. Hase Y, Dunstan P, Temperini M. Raman active normal vibrations of lanthanide oxychlorides. Spectrochim Acta Part A Mol Spectrosc. 1981;37:597–9. https://doi.org/10.1016/0584-8(81)80055-4.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.